

Lesson 30 — Design Challenge — AI Pose Story Circle

Matches LEGO Teacher Portal lesson: “Stop and Go Parade”.

Objective (I can...): I can program events with AI pose blocks so a pose makes my build do something in a little story. **CS concept:** Programming events with AI pose blocks (applying pose recognition with events and data) · **Builds on:** Lessons 26–29 and the whole course. *Honest framing: kids use ready-made pose blocks — they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved. Also serves as flex/catch-up — use today to finish and polish anything from the AI unit.*

AI Connection (2 min, read aloud): Today your build senses, collects data, and reacts to a pose that a ready-made AI block recognizes — every unit of the year working together. *Ask the group:* “Which idea from this year did your story use the most?”

Materials: LEGO CS & AI kit + device per pair with camera, plan mats, story props. **Model already built** (AI & Data model from Lesson 26) — the 25 min is plan + code + play, not build-from-scratch.

60-minute flow - Warm-up (10): Story spark — pairs pitch a one-sentence story where striking a pose changes what happens next. *(Theme: AI pose show.)* - **Concept + demo (8):** Challenge: program events with the pose blocks so a recognized pose drives the model and the story (arms up → the rocket launches). Coach shows a short plan connecting a pose to a story beat. Restate privacy points. - **Build/Code (25):** Pairs plan and code a little interactive story where at least one pose event drives the model, plus (if they can) a count or a loop, on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second pose that leads to a different ending. - **Share + tidy-up (10): AI pose show** — each pair performs their pose story while the coach records a short video clip for the parent note. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator directs the story and strikes the poses; swap at the wiggle break. *(Trio rotates plan/code/perform.)*

Differentiation: *Support* — one pose, one story beat, partial plan. *Extension / fast-finisher* — two poses with two endings and a count. *Reteach* — reuse their Lesson 28 program and add just one story reaction.

Exit check: Pair shows a working pose story where a pose event makes the build do something, with a matching plan. **Progress layer:** Send home the **Unit 6 (course finale) parent note** (“your child can collect data and use ready-made AI pose blocks to build a machine that senses, repeats, and responds — and can explain how a computer learns to recognize things from examples”) with the **build photo** — the year’s capstone. *Keep the framing honest: your child explored how AI works and used AI pose blocks; training a classifier of their own is next, in the 3–5 band.* **At-home (optional):** Draw a 3-picture pose story; mark which pose changes the ending. ~5 min.

Standards Crosswalk

Aligned to the **CSTA K-12 CS Standards** — primarily **Algorithms & Programming (1A-AP)**, with connections to **Data & Analysis (1A-DA)** and **Impacts of Computing (1A-IC)**. These are the Level 1A (grades K-2) codes, mapped to the six real units. This is a targeted map, not a claim of full coverage. *Maps to the Virginia CS Standards of Learning, grades K-2 (confirm exact SOL numbers).*

Unit (Lessons)	Skills taught	CSTA 1A codes targeted
1 • Basics (1–5)	Collaboration routines (Driver/Navigator), modeling everyday processes as algorithms, breaking a job into little steps, saying the plan first	1A-AP-08 (model daily processes as precise step-by-step algorithms); 1A-AP-10 (develop programs with sequences); 1A-AP-11 (decompose steps into a precise sequence)
2 • Hardware (6–10)	How the motor works (start/stop, speed, direction), how the color sensor reads, motor and sensor working together, using color to stand for information	1A-AP-10 (programs that control hardware); 1A-AP-09 (use color/symbols to represent information); 1A-AP-14 (test and fix motor behavior)
3 • Events (11–15)	Events that cause something to happen, traffic-signal color events, creating events, hardware responding to events, programming a motor with events	1A-AP-10 (events and control in programs); 1A-AP-08 (precise step-by-step behavior); 1A-AP-11 (decompose a triggered behavior)
4 • Sequences (16–20)	Ordered steps, debugging/developing a sequence, a sequence that solves a problem, picture pseudocode, a story in order	1A-AP-11 (order steps into a precise sequence); 1A-AP-10 (develop sequenced programs); 1A-AP-12 (develop plans that describe a program's sequence); 1A-AP-14 (debug)
5 • Loops (21–25)	Repeat loops, developing a loop body, repeating + collecting a simple count, creating loops, pattern builds	1A-AP-10 (loops/repetition in programs); 1A-AP-14 (debug/Fix-It); 1A-DA-06 (collect and present simple data — the repeat count)
6 • AI & Data (26–30)	Collecting/describing/representing data, modifying a program that collects & visualizes data, using pretrained AI pose blocks, modifying pose programs, programming events with pose blocks, why computers guess wrong (fairness)	1A-DA-05 (store, copy, and sort information to make it useful); 1A-DA-06 (collect and present data — tallies and simple charts); 1A-IC-16 (how people use computing / being fair); 1A-AP-10
Pair programming (every lesson)	Driver/Navigator roles, swapping, taking turns, following and giving instructions	1A-AP-12 (develop/follow plans together); collaboration across all lessons
Cross-cutting: Fix-It and debug mini-tasks (L10 catch-ups, L17, L19, L22) reinforce 1A-AP-14 ; the "two ways to do it" and design-challenge share-outs reinforce 1A-AP-12 ; the AI-impacts prompts (L8 self-driving cars, L14 always-listening speakers, and the L29 fairness discussion) reinforce 1A-IC-16 ; the data-collection and tally work in Units 5–6 reinforces 1A-DA-05 / 1A-DA-06 .		
Honesty note for coaches and families: Throughout Unit 6, K–2 students explore how AI works — they collect and show simple data, feed a coach-run Teachable Machine demo examples, and use ready-made (pretrained) AI pose-recognition blocks to drive a build. They do not build or train their own AI model . Training a custom classifier is a grade 3–5 skill and lives in the Builders and Engineers bands. Keep that promise accurate on every flyer, parent note, and share-out.		